GAMMA LONG

Ten stocks, chosen monthly, from the Nifty 100, Midcap 100 and Smallcap 100

Finesse x Citadel Challenge, Round 2
Submission, 31 August 2026
Backtest 1 January 2021 to 31 December 2025

NET PROFIT	CAGR	MAX DRAWDOWN	SHARPE	ACCURACY	VS RANDOM
Rs 5.75 cr	**46.6%**	**20.1%**	**2.10**	**62.3%**	**99.4th**

Ten equally weighted stocks, rebalanced monthly. A name must sit in one of the three permitted indices on that date, clear a liquidity floor of Rs 5 crore of daily traded value, and be better run than the median company in its own sector. Survivors are ranked on risk adjusted momentum blended equally with proximity to the 52 week high. Rs 1 crore became Rs 6.75 crore after costs, at a worst drawdown of 20.1%.

1. THE PROBLEM HAS TWO ANSWERS

Ten names at most, drawn from roughly 300 index members, Rs 1 crore of capital, 0.1% on every transaction. Teams are ranked on five year net profit, and the submitted model is then run forward on unseen 2026 data. Those two tests reward opposite behaviour, so the rule was fixed from theory first and the backtest was allowed only to confirm or reject it.

2. WHY THE MODEL LOOKS LIKE THIS

Every rule names a counterparty. If we cannot say who is on the other side and why they keep losing, we have a backtest rather than an edge. For momentum it is the investor who sells a winner early to book the gain, the disposition effect of Shefrin and Statman (1985).

Momentum is a fork, not a chain. Past return does not cause future return; both are traces of the same unobserved repricing (Exhibit 1). So the signal is measured three times, at twelve, six and three months, and averaged. On 2007 to 2025 that average beats every leg it is built from.

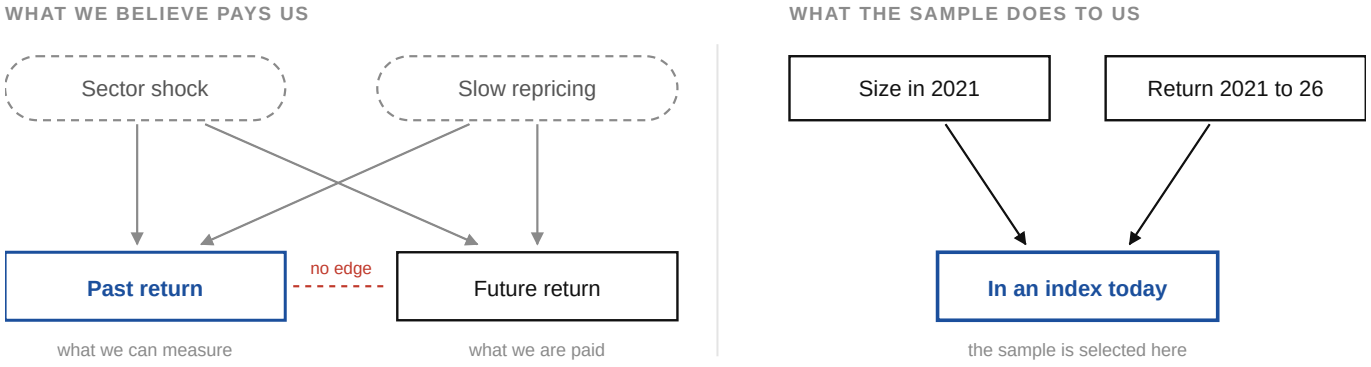
Sector shocks confound it. In a simulation with zero stock level persistence built in, the naive past against future rank correlation is +0.34, with a t statistic of 26. Demeaned by sector it is 0.008. Every comparison is therefore made inside a sector.

The universe itself is a trap. Membership today is a collider fed by both 2021 size and subsequent return, and the whole sample is selected on it. Size and illiquidity were excluded before any data was read: the measured size information coefficient is +9.0% pooled and +24% to +26% inside one bucket. The Rs 5 crore floor cuts the artefact to 9.0%, but nothing cures it.

EXHIBIT 1

The signal is a fork off something unobserved, and the sample is selected on a collider

Causal graph of the model's claims. Dashed outline: not observed. Blue: what the model conditions on



Source: Shefrin and Statman (1985), Journal of Finance; Pearl, Causality (2009); López de Prado, Causal Factor Investing (2023); Gamma Long analysis

3. DATA

Prices. Daily closes for 548 symbols, 2005 to 2026, from the file supplied with the problem, extended with Yahoo Finance for older history and volume. No per share line is read, so every accounting test is a ratio of aggregates and is invariant to splits.

Membership is point in time. Constituent lists are rebuilt for each rebalance date by walking the NSE reconstitution log backwards. 478 companies pass through the three indices and 308 are present for only part of the window (Exhibit 3).

Fundamentals. Annual statements for 484 symbols from screener.in. A financial year becomes visible only six months after it ends, so the accounts behind a decision are 244 days old at the median. Shortening that lag makes results worse, which is the cleanest evidence that no look ahead has crept in.

Benchmarks and tools. No free total return series exists, so each index is compounded at a constant dividend yield, 1.40% and 1.30%. Python with pandas, numpy and scipy. No machine learning and no optimiser.

4. HOW A STOCK ENTERS THE BOOK

Four tests, every month. Of roughly 300 index members, 284 clear history, price and the traded value floor, and 112 pass the quality test (Exhibit 2). The 17 names a month with no usable data are carried unscored rather than deleted.

Equal weights, because the score cannot size. An information coefficient near 11% is strong enough to say which names to own and far too weak to say how much of each.

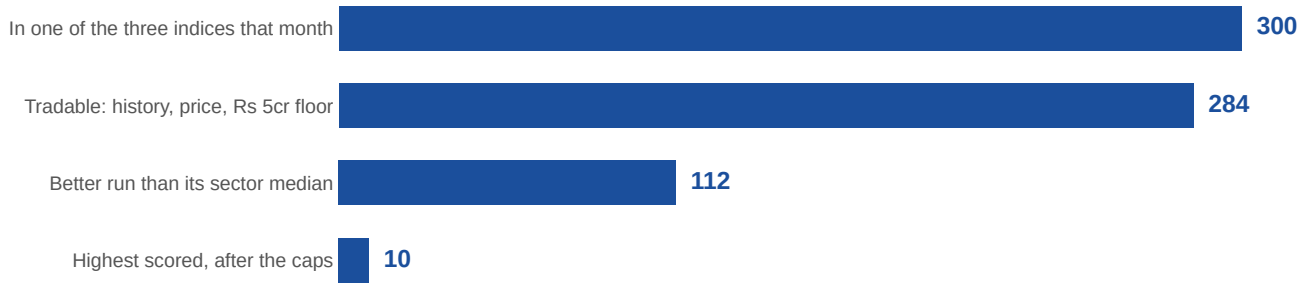
Three caps and one exit rule. A held name stays while it remains in the top twenty. At most three names per industry, and five from the Smallcap 100. A name that leaves the universe or the floor is sold at that rebalance, so the buffer never overrides eligibility.

Assumptions. Trades execute at the month end close and pay 0.1% each way, Rs 16.3 lakh over five years. Impact, slippage and taxes are not modelled. Always fully invested: no cash, no leverage, no shorting.

EXHIBIT 2

Two tests of tradability and one of quality turn roughly 300 index members into a book of ten

Average number of names surviving each stage, per rebalance date, 60 rebalances

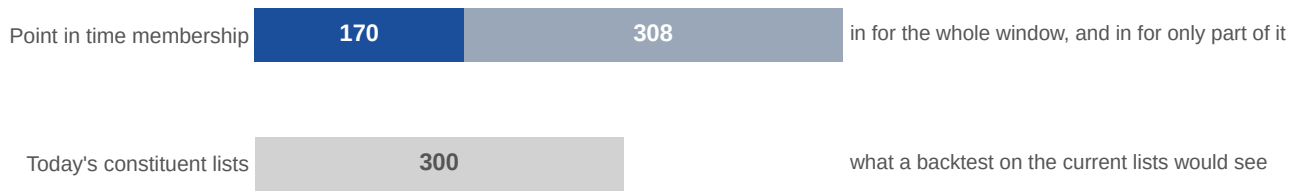


Source: NSE Indices; Yahoo Finance; screener.in; Gamma Long analysis

EXHIBIT 3

Rebuilding membership through time recovers 308 companies the current lists never show

Distinct companies eligible at some point between January 2021 and December 2025



Source: NSE Indices reconstitution circulars, March 2020 onwards; Gamma Long analysis

5. HOW THE MODEL WAS BUILT, AND WHAT WAS THROWN AWAY

Five increments, each with a test written down before it was run. Exhibit 4 lists them. The evidence sits outside the scoring window wherever the data allows, and only two of the five move terminal wealth by more than the resolution limit of section 7.

One rule was removed rather than added. An earlier version also required a valuation test. Over 2016 to 2025 quality alone compounded at 31.4% against 25.9% for value plus quality, and it won in the 2016 to 2020 half. The value gate came out on that evidence.

Thirteen further candidates were rejected (Exhibit 5). The most instructive cleared the raised evidence bar at signal level, with a t statistic of 4.15, and was still not adopted: at book level it landed at the 72nd percentile of pure noise. Clearing a statistical bar and being measurable in ten names are different things.

EXHIBIT 4

Each step had to pass a test stated in advance, on evidence outside the scoring window wherever possible

The five increments that make up the score, in the order they were admitted

STEP	MECHANISM	TEST WRITTEN DOWN IN ADVANCE	MEASURED	WINDOW
1. Twelve month momentum, last month skipped	Slow repricing; sellers who book gains early	Positive information coefficient outside 2021 to 2025	IC 6.7%, t 4.5	2007 to 2025
2. Three horizons averaged	One latent measured three times; averaging cancels noise	The average must beat every leg, or the legs are independent	8.2% against 6.7%, 7.7%, 6.1%	2007 to 2025
3. Each leg divided by its own volatility	Pays for risk taken, not risk borne	Gain must come from reallocating between volatility levels, not within	Nothing within quintiles; all of it between	2007 to 2025
4. 52 week high proximity, half the score	Anchoring: buyers hesitate near the high, then capitulate	Must raise the information coefficient over momentum alone	11.2% to 12.9%, t 3.5 to 4.1	2021 to 2025
5. Quality gate on sector medians	Momentum with no solvency check buys junk rallies	Must win in 2016 to 2020, which does not overlap	2.25x against 1.89x	2016 to 2020

Source: Jegadeesh and Titman (1993) and George and Hwang (2004), Journal of Finance; Barroso and Santa-Clara (2015) and Novy-Marx (2013), Journal of Financial Economics; Agarwalla, Jacob and Varma (2017), Vikalpa 42(4); Gamma Long analysis

EXHIBIT 5

Thirteen candidates were tested against the same sixty months and none was adopted

Selected rejections, with the reason each failed

CANDIDATE	WHAT IT CLAIMED	WHY IT WAS NOT ADOPTED
Earnings announcement drift	Prices adjust slowly to results, so recent surprise predicts return	Cleared the raised bar at signal level, IC 16.9% and t 4.15, the only candidate to do so. At book level the result sat at the 72nd percentile of pure noise, so the effect is unmeasurable in ten names rather than absent
52 week high recency	How recently the high was set adds information beyond how close price is to it	Correlated 0.69 with the leg already in the score, and the information coefficient moved from 16.34% to 16.33%. The same measurement twice
Frog in the pan	Gradual price paths are underreacted to more than jumpy ones	Standalone information coefficient of −0.4%, t of −0.12; as a tilt it lowered the score. Rejected twice, once as a gate and once as a tilt
Momentum seasonality	Returns repeat at annual lags, so on cycle months carry the effect	Killed by its own falsifier: off cycle lags were stronger than on cycle ones, making it long horizon reversal rather than a calendar effect
Maximum daily return	Lottery like stocks are overpriced, so screening them out should add return	Neither the international sign nor the published Indian sign replicates here, and every bucket is insignificant. A screen would have been a coin flip

Source: Bhootra and Hur (2013), Journal of Banking and Finance; Da, Gurun and Warachka (2014), Review of Financial Studies; Heston and Sadka (2008) and Bali, Cakici and Whitelaw (2011), Journal of Financial Economics; Ali, Hasan and Östermark (2020), International Review of Economics and Finance 70; Gamma Long analysis

6. RESULTS

Rs 1 crore became Rs 6.75 crore after costs. That is 46.6% compounded against 17.2% for the Nifty 500 total return index, at a worst fall of 20.1%. The Nifty 500 is the benchmark because it is the only widely quoted index spanning all three permitted buckets.

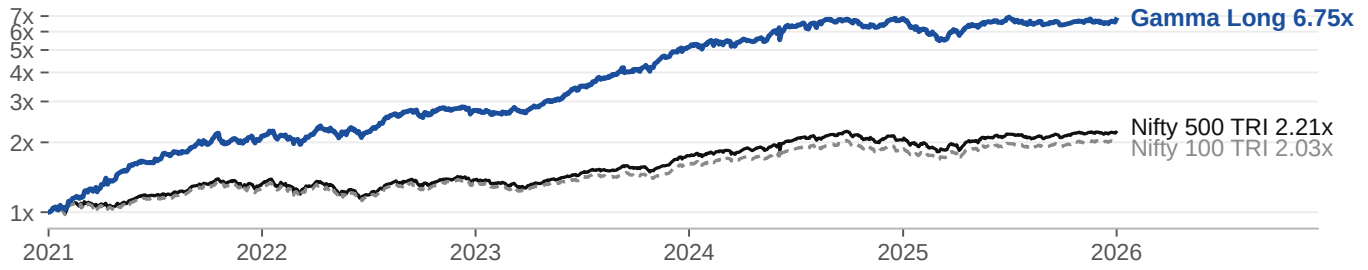
Beating the index is close to automatic and we claim no credit for it. The universe is tilted to mid and small caps and the benchmark is not, so 82% of random ten stock baskets also beat it. Against those baskets Gamma Long sits at the 99.4th percentile of 5,000 draws, with a beta of 1.03 and alpha of 25.9%, t statistic 4.06.

Trades are counted strictly. A trade is a round trip from entry to full exit, so the trims an equal weight rebalance generates belong to the position rather than counting separately. On that basis 62.3% of 191 round trips made money; counting every sell line would report 79.6%.

EXHIBIT 6

Rs 1 crore compounded to Rs 6.75 crore, against Rs 2.21 crore for the Nifty 500 total return index

Growth of Rs 1 crore after 0.1% transaction costs, log scale (x)

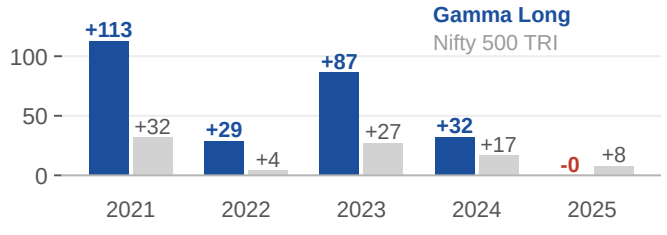


Source: NSE Indices; Yahoo Finance; Gamma Long analysis

EXHIBIT 7

Four years of large outperformance, and one year of falling behind

Calendar year total return, per cent, after costs

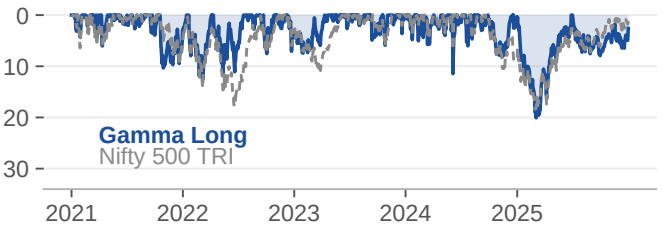


Source: NSE Indices; Yahoo Finance; Gamma Long analysis

EXHIBIT 8

The deepest fall was 20.1%, against 18.4% for the benchmark

Drawdown from the running peak, per cent, daily marks



Source: NSE Indices; Yahoo Finance; Gamma Long analysis

EXHIBIT 9

Every metric the brief asks for, on the strict definition of a trade

1 January 2021 to 31 December 2025, after costs

PERFORMANCE	GAMMA LONG	NIFTY 500 TRI
Final value of Rs 1 crore	**Rs 6.75 cr**	Rs 2.21 cr
Total net profit	**Rs 5.75 cr**	Rs 1.21 cr
Absolute return	**+575.2%**	+121.1%
Annualised return	**46.6%**	17.2%
Maximum drawdown	20.1%	18.4%
Annualised volatility	19.4%	14.4%
Sharpe ratio	**2.10**	1.20
Return over max drawdown	2.32	0.94
Beta, and alpha t statistic	1.03, t 4.06	1.00
Out of sample, H1 2026	**+6.3%**	**–**

TRADING	VALUE	BASIS
Accuracy, profitable trades	**62.3%**	191 round trips
Gain to loss ratio	**2.01**	Rs 6.62 / 3.30 lakh
Transactions	791	60 rebalances
Transactions per name	6.8	116 names held
Value traded	Rs 163.0 cr	both sides
Transaction cost paid	Rs 16.3 lakh	at 0.1%
Monthly turnover, one way	66.8%	weight changes
Median holding period	2.0 months	mean 3.1
Positive months	73.3%	of 60
Best and worst month	**+14.8% / 9.7%**	**60 months**

Source: NSE Indices; Yahoo Finance; Gamma Long analysis. Every figure recalculates live from the trade blotter in the submitted workbook

7. WHAT IT HELD, WHAT COULD BE LUCK, AND WHAT IS WRONG WITH IT

The rule never changed and the portfolio changed completely. Exhibit 10 shows one name of overlap between consecutive year ends, and none between 2024 and 2025. Risk is managed inside the book: equal weights bound single name risk and three names per industry bound sector risk.

A ten stock book has a wide resolution limit. Perturbing the score with information free noise and running 200 backtests, 90% of the reshuffles land between 3.42x and 4.80x (Exhibit 11). Nine points of compound return is noise at this size, so the honest forward expectation is that distribution's median, near 34%.

Every parameter was moved one at a time. Across 72 variations on 17 axes (Exhibit 12), compound return ranges from 20.7% to 50.0%. Twenty six fall more than a noise band below the submitted book and none is better.

Limitations. The survivorship collider is unfixable and is stated rather than solved. Fundamental values are the vendor's current view of each past year. The quality test is invalid inside financial services, where a bank's assets are its loan book: correcting it costs 3.5 points and improves drawdown to 17.7%.

EXHIBIT 10

One name of overlap between consecutive year ends, and none at all between 2024 and 2025

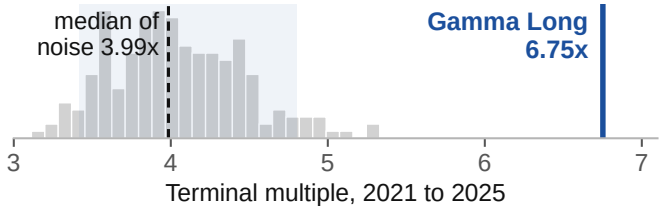
The ten equally weighted holdings carried into each year end

2021	2022	2023	2024	2025
Bajaj Finserv	ITC	Power Finance	Sun Pharmaceutical	Bharat Electronics
Bajaj Holdings	Cummins India	REC	HCL Technologies	Eicher Motors
Titan Company	Coal India	Sun TV Network	Persistent Systems	Maruti Suzuki
Tata Elxsi	Cipla	ONGC	Coforge	Cummins India
Bharat Electronics	ICICI Bank	Coal India	National Aluminium	Indian Bank
Escorts Kubota	Hindustan Aeronautics	HCL Technologies	Bajaj Holdings	Bharat Petroleum
Carborundum Universal	Titan Company	BSE	CDSL	Muthoot Finance
Tanla Platforms	Varun Beverages	Bajaj Auto	ICICI Bank	Hero MotoCorp
Page Industries	Bharat Dynamics	Torrent Power	Oberoi Realty	Shriram Finance
CDSL	IIFL Finance	NMDC	Torrent Pharmaceuticals	Bharti Airtel
3 large, 4 mid, 3 small	**6 large, 3 mid, 1 small**	**3 large, 5 mid, 2 small**	**5 large, 3 mid, 2 small**	**6 large, 4 mid, 0 small**

Source: NSE Indices; Yahoo Finance; Gamma Long analysis

EXHIBIT 11

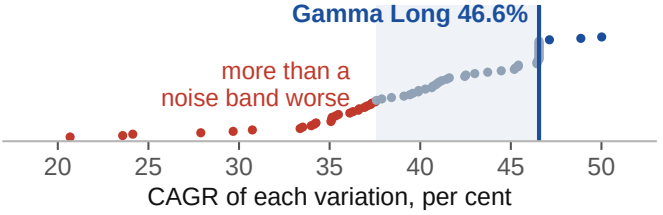
Noise alone spans 3.42x to 4.80x, so nine points of return is the resolution limit



Source: Gamma Long analysis

EXHIBIT 12

Of 72 variations, 26 are more than a noise band worse and none is better

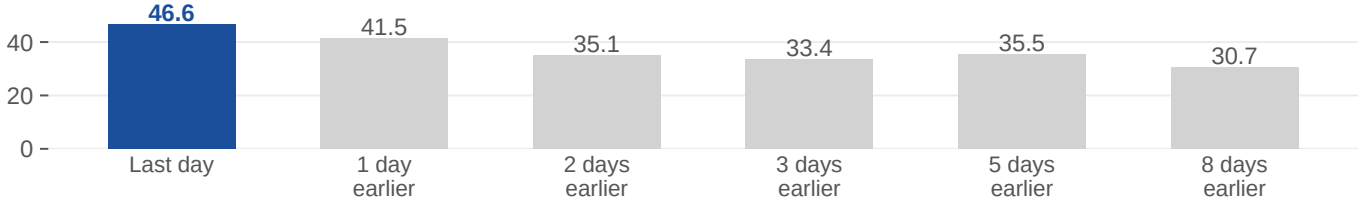


Source: Gamma Long analysis

EXHIBIT 13

Rebalancing a few days earlier costs up to sixteen points, which is luck rather than skill

Compound annual return by rebalance date, per cent



Source: Yahoo Finance; Gamma Long analysis